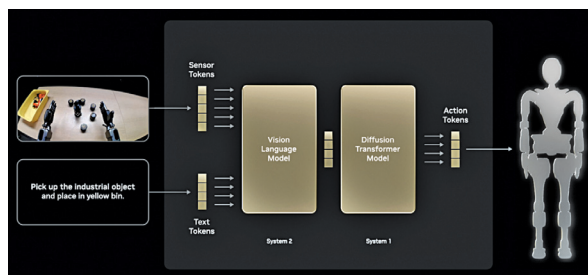


Open humanoid robots

Nvidia's open humanoid robot reference design simplifies humanoid



robot development by combining AI models, computing hardware, simulation tools, and robotic hardware into a single development platform that supports building, training, and deploying humanoid robots. At its core is Isaac GR00T, an open foundation model based on a vision-language-action architecture that enables robots to understand natural language, interpret their surroundings, and generate appropriate actions through separate systems for real-time motor control and higher-level reasoning and planning.

The platform also integrates edge computing, a humanoid robot platform, and dexterous robotic hands to support perception, planning, real-time control, object manipulation, and tool handling. To further accelerate development, you can use the platform's simulation tools to train and test your robots in virtual environments using synthetic data.

Applications. The reference design aims to simplify humanoid robot development and accelerate deployment across industries.

Link. <https://www.electronics-foru.com/electronics-projects/open-reference-design-for-humanoid-robots>

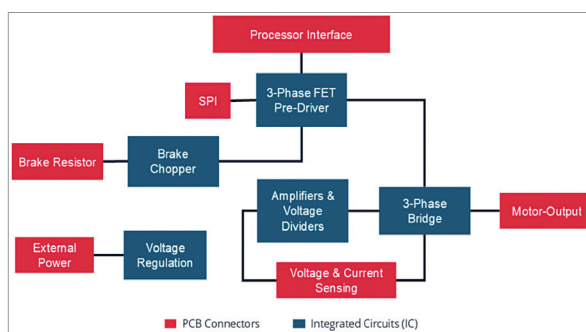
OEM Brand. Nvidia

Humanoid major joint motor control

Molex has released a major joint motor control reference design to help you develop power and signal connectivity for humanoid robot joints such as the shoulders, hips, and knees. It addresses the challenges of major joint motors, which require

high-current power distribution and fast switching for precise position and torque control, often creating issues

related to power integrity, thermal management, and connector reliability in compact robotic systems. To support these requirements, the design uses high-current power connectors with terminals rated up to 50A, secure locking mechanisms, and stable contact interfaces that ensure reliable power delivery



to motor drives, braking circuits, and sensing systems even under continuous motion and vibration.

For lower-power signal connections, it incorporates compact signal connectors available in multiple circuit configurations with a 1.25mm pitch and positive locking features that help maintain reliable electrical connections, prevent incorrect mating during assembly, and support dense PCB layouts.

Applications. The reference design is suitable for humanoid robot shoulder, hip, and knee joints.

Link. <https://www.electronicsforu.com/electronics-projects/humanoid-major-joint-motor-control-reference-design>

OEM Brand. Molex

GaN-based motor drive for humanoid

Efficient Power Conversion Corporation (EPC) has introduced a reference design for motor drives in humanoid robot joints that integrates gallium nitride (GaN) power technology into a compact, lightweight platform.

The design supports three-phase BLDC motors while combining the GaN power stage, current and voltage sensing, a magnetic rotor position encoder, a microcontroller, RS-485 communication, and onboard power supplies on a single circular board.

By using GaN technology, the reference design enables higher switching frequencies, lower power losses, and the use of compact multilayer ceramic capacitors instead of bulky electrolytic capacitors, reducing inverter size by 66% compared with conventional silicon-based designs

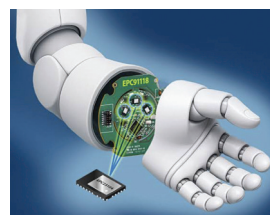
while improving reliability.

Applications. This reference design is well-suited for high-performance motion control in humanoid robots and other compact robotic systems.

Link. <https://www.electronicsforu.com/electronics-projects/compact-gan-based-inverter-reference-design>

OEM Brand. Efficient Power Conversion Corporation

These reference designs provide



practical starting points for solving different motion-control

challenges, from driving individual finger joints and major limb actuators to building complete humanoid platforms with AI, simulation, and advanced power electronics. By using these evaluation-ready designs, you can reduce development time, validate your motor-control architecture more quickly, and focus on improving robot performance instead of building every subsystem from scratch. **EFY**

Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.